

Explain this product to a real company.

A complete briefing on the Virtual EMC Pre-Compliance Advisor: what electromagnetic compatibility is, why elevator drives emit noise, how the physics simulator and the machine-learning models work, what every number on the screen actually means, and how to present the tool without overselling it.

Internal study document · 2026-09-16 · artifact version 2.0.0 · written for a reader who does not already know EMC, power electronics, or machine learning.

READ THIS BEFORE YOU WALK INTO THE ROOM

This tool estimates conducted-EMC **risk from a physics simulation**. It does not predict EN 12016 certification. It has never been fitted to chamber measurements. The limit curve is a documented assumption, not the copyrighted standard table. Direction and size of a design change are the reliable output. An absolute LOW / MODERATE / HIGH badge is only as good as that assumed curve. If you remember one sentence, remember that one.

How to use this document

- **Ten minutes before a meeting:** memorise the one-minute pitch on the next page, then skip to Part V (what to say, what never to say, likely questions).
- **One evening of study:** read Parts I, III and V in order. That is enough to demo the product and survive a mixed audience of managers and engineers.
- **If an EMC engineer will be in the room:** also read Parts II and IV. They will ask about the limit line, common-mode current, monotone constraints, and the 99% accuracy trap.
- **If you only need a leave-behind:** print this PDF. Do not print a LOW-risk assessment PDF and call it a certificate.

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THE ONE-MINUTE PITCH

Memorise this. Then sit down.

We built a virtual pre-compliance advisor for conducted emissions on elevator variable-frequency drives. You give it six design numbers — switching frequency, edge speed, motor-cable length, shield quality, load current, and PWM strategy. It synthesises the inverter waveform, the common-mode current on the motor cable, and the spectrum a CISPR-style receiver would see. A physics-constrained model then returns a three-tier risk flag, an explanation of *why this design sits where it does*, and a separate list of *what still has room to change*.

It is not a lab. It is not a certificate. The limit line is a synthetic CISPR-shaped envelope, because we cannot redistribute the EN 12016 tables. Every accuracy figure is agreement with our own simulator, not with hardware. What it is good for is the question that normally waits for a chamber booking: is this cable run, this carrier, and this shield going to be a problem — and which knob still moves the answer.

SAY THIS

“Treat this as an early-design risk indicator. Use it to compare options and to walk into the lab less blind. Accredited measurement remains the only certification path.”

NEVER SAY THIS

“It certifies the drive.” “It predicts EN 12016.” “It is 99% accurate.” “The ensemble plus/minus is lab uncertainty.” “You can skip the chamber.” Any of those sentences ends the meeting.

PART I · THE SITUATION

What problem this is even for

An elevator drive is a box of power electronics. It takes building mains and turns it into a controlled waveform that runs a traction machine. To do that, transistors switch hundreds of volts on and off thousands of times per second. Every switch is a tiny radio transmitter. The motor cable in the shaft is the antenna. Other equipment in the building — and the lift’s own safety electronics — are the victims.

EMC means electromagnetic compatibility: the drive must not pollute the electrical environment enough to upset neighbours, and it must keep working when the environment is noisy. For lifts and escalators in Europe the product-family standard people name in this context is **EN 12016** (immunity) together with the emission limits the sector actually tests against, which are CISPR / EN 55011 shaped. Getting that wrong late is expensive. A shielded cable, a common-mode choke, or a slower semiconductor can be a cheap line on a drawing and a painful retrofit on a site.

The honest bottleneck is the **EMC chamber**. A proper conducted-emission test needs a Line Impedance Stabilisation Network (LISN), an EMI receiver, and time on a controlled site. You do not get a booking every time a firmware engineer wants to try 12 kHz instead of 8 kHz. So teams either over-design (cost, heat, acoustics) or under-design (fail the lab, slip the programme).

This tool sits in that gap. It answers, in seconds, a *relative* question: given this design, relative to a documented assumed limit, where is the risk, and which parameter is carrying it. It does not replace the chamber. It makes the chamber a confirmation step instead of a discovery step.

EMC in plain language

Electricity is supposed to stay in wires. In real life, fast voltage changes push current through parasitic capacitance — the invisible capacitor between a cable and the metal of the shaft, or between a transistor and its heatsink. That current has to come home somehow. On the way home it flows through earth, through the mains, and through anything else that happens to be in the loop. A test house measures that homeward current as a voltage across a standardised 50 ohm network, in decibels relative to one microvolt (dBµV).

Two families of test exist, and this product only claims one of them:

Family	What is measured	In this tool
Conducted emissions	Noise riding on the power cable, typically 150 kHz–30 MHz	Yes — this is the whole product
Radiated emissions	Noise leaving as a radio wave, typically 30 MHz–1 GHz+	Out of scope
Immunity	Whether the drive still works when someone else's noise hits it	Out of scope
Power quality / THD	Distortion of motor and input current at 50 Hz harmonics	Shown beside the risk score, not mixed into it

Conducted and radiated are related — the same edges cause both — but a good conducted result is not a radiated pass. Do not let anyone collapse them in the meeting.

What a lab actually does

A conducted-emission test does not look at a pretty oscilloscope trace. An EMI receiver dwells on each frequency with a legally defined **resolution bandwidth** (9 kHz in CISPR band B, which is 150 kHz to 30 MHz). It max-holds what it sees, because the inverter's duty cycle breathes over the 50 Hz output cycle and a single snapshot would lie. The result is a curve of dB μ V versus frequency, compared against a limit line.

Three sub-bands are how drive people usually talk about that curve:

Band	Range	What tends to live there
A	150–500 kHz	Switching harmonics, the bulk of the envelope
B	500 kHz–5 MHz	Still switching structure; often the tightest band here
C	5–30 MHz	Edge speed, cable resonances, shield quality

Pre-compliance is not certification

Certification is an accredited laboratory measuring a physical product against a named standard, with a report that a notified body or a customer will accept. **Pre-compliance** is everything you do before that so you are not paying for surprises. Near-field probes, a LISN on a bench, a simulator, a risk model — all pre-compliance. Useful. Not the same legal object.

EMC Advisor is labelled, in the UI and on every generated PDF, as a **Virtual EMC Pre-Compliance Report**. The word “certificate” was deliberately not used. There is no accreditation mark, no signature block, and no statement of conformity. If a slide deck behind you says “AI certification for EN 12016”, take the slide down before you speak.

PART II · THE PHYSICS

From a transistor to a spectrum

You do not need a power-electronics degree. You need six pictures in your head, because every knob on the screen is one of these pictures with a number attached.

1. A VFD is a fast switch, not a dimmer

A variable-frequency drive does not turn the motor voltage “partly on”. It slams the DC link (here modelled as 565 V, a rectified 400 V three-phase supply) onto each motor phase, then slams it off again, thousands of times per second. The average of those pulses is the smooth voltage the motor wanted. The pulses themselves are the EMC problem. The pulse repetition rate is the **switching frequency** (3–16 kHz in this tool, the range of ordinary elevator IGBT and SiC carriers). Raise it and the motor is quieter and smoother. Raise it and you also put more switching edges into the 150 kHz–30 MHz measurement window, so the spectrum rises.

2. The edges, not the pulses, set the high-frequency noise

A perfect square pulse has energy that falls 20 dB per decade of frequency, then 40 dB per decade after a corner set by the rise time. That is the classical trapezoidal EMI model. Faster edges (higher dv/dt , in volts per microsecond) push that corner up. Silicon carbide devices are efficient partly because they switch very fast — and they are noisy in band C for the same reason. This tool lets dv/dt run from 500 to 10 000 V/ μ s. A 9.5 kV/ μ s SiC edge is a different EMC object from a 1.2 kV/ μ s freight-drive IGBT.

3. Common mode is the actual villain

The three inverter legs are called A, B and C. The voltage that drives the motor *between* those legs is differential and mostly stays in the motor. The voltage that drives current *into the earth* is the average of the three: $v_{cm} = (v_a + v_b + v_c) / 3$. For a two-level inverter that average is a staircase that steps six times per carrier period. That staircase is the conducted-emission source. This is why a prettier line-to-line PWM picture can still be a disaster on the LISN.

Current then follows $i = C \cdot dv/dt$. The C is parasitic capacitance from the motor cable to the shaft (modelled here as 100 pF per metre). Longer cable, larger C, more current. That is why an 85 m ultra-rise shaft cable is a different risk from a 12 m machine-room-less run, even if the inverter is “the same”.

4. The cable is also a radio resonator

A long cable is a transmission line. It rings at a quarter-wave resonance $f \approx v / (4 \times \text{length})$ and at the odd harmonics 3f, 5f. If you only modelled the first resonance, lengthening the cable would sometimes slide the peak *out* of the 5–30 MHz band and the model would stupidly report that longer cables are quieter there. The simulator therefore keeps three resonant modes, so more cable still means more coupling, which is the physics the machine-learning constraints later enforce.

5. A shield is an attenuator, not a magic zero

Shielding quality in the UI is a 0–1 slider that folds together cable screen, 360-degree termination, and any common-mode choke. Zero is unscreened. One is a well-terminated screen plus a choke, modelled as up to 26 dB of insertion loss and a high-frequency roll-off. Better shielding can only reduce the emission in this model. It cannot invent energy. That sign is later locked into the trees so a noisy corner of training data cannot learn the opposite.

6. PWM strategy changes how often the villain steps

Four modulation strategies are modelled. The number that enters the model is not a rank. It is a measured common-mode penalty in dB, relative to ordinary sinusoidal PWM, averaged in the simulator itself:

Strategy	Penalty vs SPWM	Why
Sinusoidal PWM (SPWM)	0 dB	Reference. All three legs switch every carrier period.
Space-vector (SVPWM)	+1.0 dB	Same switch count, but zero-sequence injection adds common-mode on purpose (the usual DC-link utilisation trade-off).

Strategy	Penalty vs SPWM	Why
Randomised carrier	+1.0 dB	Does not reduce energy. Against a 9 kHz max-hold receiver at 3–16 kHz carriers, extra variance gives the detector more peaks to catch.
Discontinuous (DPWM)	–2.5 dB	Clamps one leg to a rail, removing about a third of commutations. Fewer impulses per second, lower measured lines.

Two limitations you should volunteer before you are asked. Real drives sometimes see DPWM worse at very low frequency because of extra zero-sequence content; that content lives around 150 Hz, three orders of magnitude below this measurement floor, so the model does not reproduce it. And randomised PWM is often sold as a 5–10 dB win; in *this* measurement it is slightly worse, because CISPR band B already cannot resolve the 3–16 kHz line structure you would be spreading. Randomisation starts to pay at much higher carriers (100 kHz-class SiC), which are outside this tool's range.

How the simulator copies a receiver

A 328 microsecond snapshot at 200 million samples per second can see 30 MHz, but it cannot see a 50 Hz motor cycle. If you analysed one snapshot you would be measuring luck: where in the fundamental you happened to land. A real receiver dwells. The simulator therefore takes **six short records** spaced across one 50 Hz period and max-holds their spectra, then integrates power across a 9 kHz resolution bandwidth. That is the curve on the dashboard.

Power quality is a different timescale on purpose. Total harmonic distortion of motor and input current needs many 50 Hz cycles, so it is computed from a cheaper 200 kS/s record. Mixing those two windows would be a mistake: the EMC capture cannot resolve mains harmonics, and the PQ capture cannot see 30 MHz. The UI keeps them in separate panels for that reason. The line reactor / input-filter slider only moves THD. It is not an XGBoost feature, so it cannot pretend to fix motor-cable EMC.

PART III · THE PRODUCT

What you are actually demoing

A three-step web app. Pick a starting profile (or a custom set of numbers). Adjust the knobs. Run an assessment. The backend is a Python API; the frontend is a React dashboard. One request simulates the physics, scores the design, explains it, and can print a matching PDF.

The pipeline, in one glance

Stage	What happens	What you can say
1. Parameters	Six design numbers plus PWM type. Input filter is PQ-only.	"These are drawing-office knobs, not lab traces."
2. simulate.py	Trapezoidal three-phase PWM, common-mode voltage, cable C and resonances, shield, LISN, plus four companion waveforms.	"A lumped physics model, not SPICE and not a measurement."
3. features.py	Receiver-emulated FFT, three-band margins, motor/input THD.	"Same shape of object a test house plots, against our curve."
4. Models	Per-band XGBoost classifier and margin regressor, monotone-constrained. Design-only models for SHAP.	"Forbidden from learning anti-physics."
5. Score	Each band margin maps through tanh to 0–100. Overall is half the mean, half the worst band.	"A design cannot hide a failing band behind two healthy ones."
6. Explain	SHAP waterfall, countermeasures, tornado, compare, history, PDF.	"Why it sits here, versus what is still worth turning."

The six knobs

Knob	Range	If you raise it
Switching frequency	3–16 kHz	Risk up (more edges per second). Acoustics usually improve.
Switching dv/dt	500–10 000 V/μs	Risk up, especially 5–30 MHz. Efficiency usually improves.

Knob	Range	If you raise it
Motor cable length	1–120 m	Risk up. Parasitic C and resonances both grow.
Shielding quality	0–1	Risk down. Screen + termination + CM choke.
Load current	5–200 A	Risk up, sub-linearly. Also raises input-current THD.
PWM strategy	four types	DPWM helps CM in this band; SVPWM and random slightly hurt.

Starting profiles

The six cards on the first screen are representative starting points, not characterisations of anyone's shipped product. Two of them are deliberately uncomfortable, so you can show a mitigation path without inventing a disaster live.

Profile	Why it exists in the demo
Standard Gearless VFD	The volume baseline: 8 kHz, 30 m, 45 A. Often the “this looks fine” case.
High-Speed Ultra-Rise	The teaching case. 85 m shaft cable. Cable length will dominate the explanation.
Compact MRL Controller	Short cable, weak screen, 16 kHz for acoustics. Different failure mode.
Freight Heavy-Duty	High current, slow edges, DPWM, basic screen.
SiC Regenerative	Very fast edges offset by a serious shield. Band C conversation.
Escalator VVVF	Short, well-routed, moderate carrier. The relaxed end of the set.

The assumed limit — the single most important caveat

EN 12016 tables are copyrighted. They are not in this repository and they are not in this PDF. The tool uses a synthetic envelope anchored on publicly described CISPR 11 Group 1 Class A quasi-peak levels (79 dB μ V around 150 kHz, 73 dB μ V above), with two tapers that are **this tool's own assumption**, including a slight tightening to 70 dB μ V at 30 MHz. Sitting below this curve is not a statement about EN 12016 conformity.

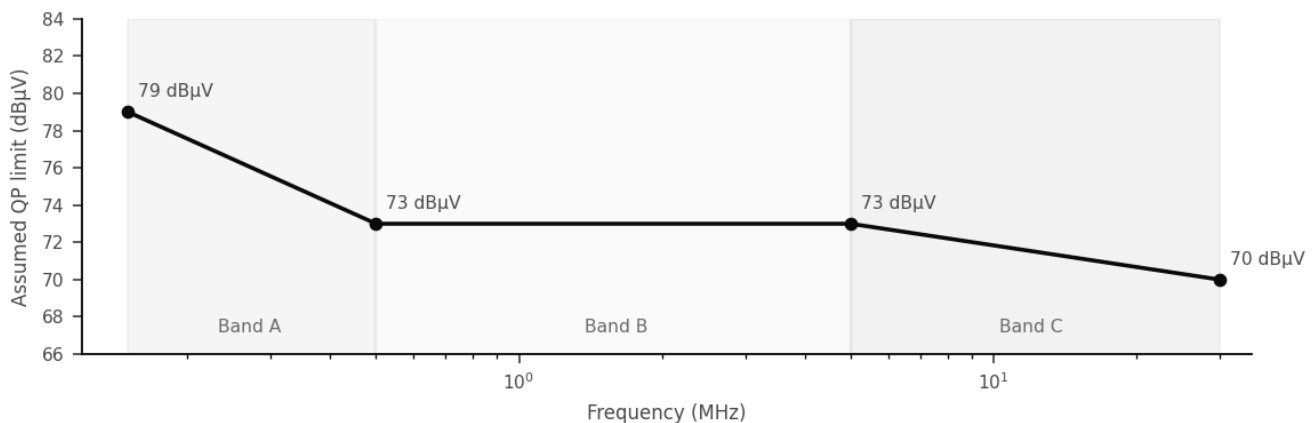


Figure. Synthetic conducted-emission envelope used as the scoring anchor. Bands A / B / C are the three ranges the models score separately.

IF THEY ONLY REMEMBER ONE CAVEAT

Absolute LOW / MODERATE / HIGH is only as trustworthy as this curve. Relative movement — shorten the cable, slow the edge, improve the screen — is the output you can defend today. Swap the anchors when you have the right to use a named table; the rest of the pipeline stays.

How a margin becomes a badge

Each band's predicted margin (limit minus emission, in dB) is mapped to a 0–100 band score with a saturating function. Zero dB of margin is exactly 50. About +8 dB is a high score; about –8 dB is a low one. The overall risk score is half the average of the three band scores and half the worst of them, so a single bad band cannot be

averaged away. Tiers: **LOW 70–100, MODERATE 40–69, HIGH 0–39**. Higher score means more simulated headroom, not a lab pass.

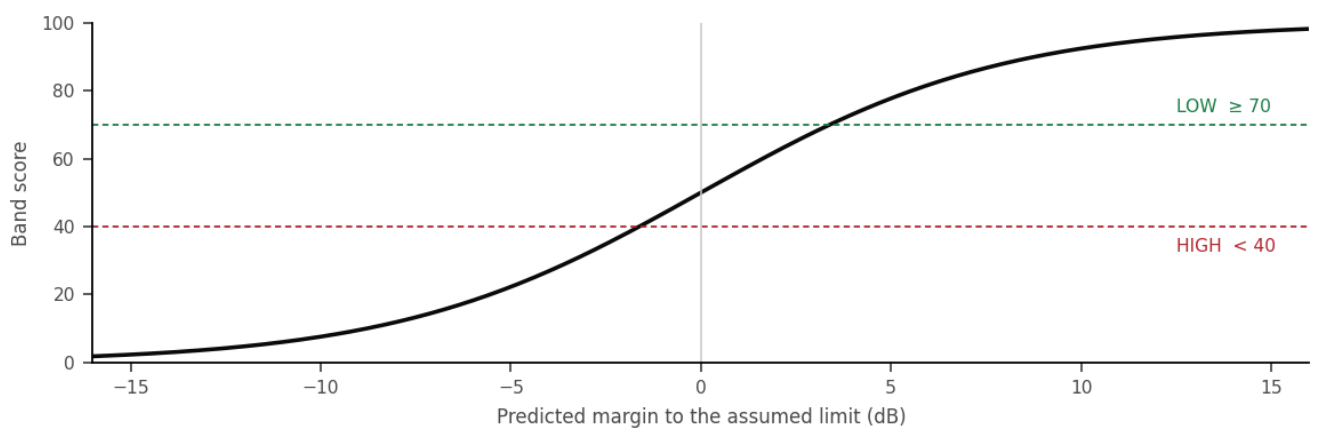


Figure. Band score versus predicted margin. The green and red lines are the LOW and HIGH cut-overs on the overall 0–100 score.

How to read the results page

Panel	Job	Do not confuse it with
Risk badge + score ±	Headline headroom against the assumed curve, plus ensemble spread.	A certification verdict, or a lab uncertainty bar.
Per-band margins	Which slice of 150 kHz–30 MHz is actually tight.	The overall badge. Always look here before you talk.
Why this margin (SHAP)	Exact attribution of the worst-band predicted margin, in dB, from the six design parameters. Largest [contribution] at the top. Red hurts margin, green helps.	A to-do list. Shielding can be the second-biggest bar because it is already good.
What to change next	Remaining headroom only. Ignores knobs already near a safe stop.	The SHAP ranking. They are allowed to disagree. The UI says so.
Sensitivity tornado	If I nudge this knob from here, how much does the score move.	A global importance ranking for all designs.
Five signals + THD	Physics you can look at: DC link, motor PWM, motor current, common-mode voltage, input current; plus motor/input THD.	The EMC limit comparison. THD is power quality.
Compare + history	Two configurations side by side; last three runs as a sparkline.	A statistical study. It is a design conversation aid.

A concrete example from the ultra-rise teaching case, worst band 500 kHz–5 MHz. The design-only margin model predicted about –12.3 dB. SHAP split that as: cable length about –18.3 dB, shielding about +10.5 dB (already helping), load current about –6.5 dB, then small negative terms from dv/dt, PWM and carrier. Countermeasures still put cable first, then pointed at dv/dt and carrier rather than at shielding — because shielding is already near the top of its range. That divergence is a feature. Explain it. Do not “fix” it in the meeting by claiming the two lists must match.

The PDF the app downloads

Same framing, same assumed limit, same disclaimer, printed in the same black-and-white language as the UI. Title: Virtual Pre-Compliance Assessment. Use it as a design note after a review. Do not staple it to a type-examination file.

PART IV · THE MODELS

Machine learning without the mystique

If you have never trained a model, this section is for you. If you have, skip to “the 99% accuracy trap” — that is the part people get wrong in front of customers.

What a model is, here

The physics simulator is the ground truth of this product. It already knows, for any design, what spectrum it would produce. Running it takes on the order of a couple of hundred milliseconds. That is fine for one assessment and painful for sweeping a design space or for interactive “what if I shorten the cable by 10 m” loops.

So the project trained a **surrogate**: a function that looks at the same numbers and returns approximately the same margins, instantly. The family used is **XGBoost** — gradient-boosted decision trees. In English: it builds hundreds of small “if cable longer than X, and shield worse than Y, then add Z dB of trouble” rules, and adds them up. Trees are a good fit because the physics is full of thresholds (resonances moving between bands) and because trees can be forced to respect a direction.

Per frequency band the tool fits two such functions: a classifier that estimates the probability the band is on the wrong side of the assumed limit, and a regressor that estimates the margin in dB. The headline score uses the margins. A third, smaller family of models sees only the six design parameters, with the spectrum hidden from them. Those **design-only** models are what the SHAP chart uses, so the explanation is about knobs a designer can turn, not about “the peak in band B was high” (which is just restating the answer).

Where the training data came from

Five thousand random designs were drawn inside the allowed ranges, simulated, and labelled with the three band margins. Split: 3 500 train / 750 validation / 750 test. There is **no hardware in that loop**. The models are students of the simulator. They can be brilliant students and still be wrong about a real shaft, because the teacher has never seen a real shaft.

The physics-informed part: monotone constraints

Ordinary machine learning will happily learn nonsense if the data wobbles. A noisy batch in which, by chance, a few long-cable designs looked quiet could teach the trees that cable length helps. That is unacceptable in a product you will show to drive engineers.

XGBoost therefore receives, for every input, a hard sign. Increasing switching frequency, dv/dt, cable length, load current, PWM penalty, or any spectral “how hot is this band” feature may only **increase** failure risk. Increasing shielding quality may only **decrease** it. For the margin regressors the signs are flipped, because more margin is safer. The constraint is structural: every split in every tree is required to move the prediction in the legal direction. It is not a post-hoc check. After training, a sweep still verified that all six design-parameter constraints hold.

HOW TO SAY THIS TO A SCEPTICAL ENGINEER

“The model is not allowed to conclude that a longer cable is quieter, or that a better shield is noisier, even if a pocket of simulated data suggests it. That is the difference between a curve-fit and a physics-informed surrogate.”

The 99% accuracy trap

On held-out simulated designs, the full 18-feature models look almost perfect: balanced accuracy around 0.98–1.00, margin errors of a few tenths of a dB. That number is not a miracle and must not be on a slide titled “AI accuracy”. The models can see **band_*_peak_dbuv** — essentially the height of the same spectrum the label was computed from. Predicting the label from the peak is close to reading the answer out of the question paper.

The honest figure is the **design-only ablation**: hide the spectrum, give the trees only the six knobs, and ask them to infer the margin. That is the job a designer actually has before anyone has measured anything. Those figures, on the same 750 test designs:

Band	Design-only balanced acc.	Design-only margin MAE
150 kHz – 500 kHz	0.964	1.06 dB
500 kHz – 5 MHz	0.972	1.23 dB
5 MHz – 30 MHz	0.939	1.88 dB

Still tight against the simulator — and still not lab accuracy. Band C is the hardest, which matches the physics: 5–30 MHz is where lumped cable and shield models are most likely to be wrong on a real fixture. If a customer later gives you chamber data, this is the band to watch.

A related negative result is kept in the methodology on purpose. Re-simulating the same design with different noise seeds moves the margin by about 0.25 dB; the model tracks that rather than averaging it away. Do not claim the ML “filters simulator noise”. It does not.

SHAP, in one paragraph

SHAP (SHapley Additive exPlanations) is a way of splitting a prediction into a sum of per-feature contributions plus a baseline. For trees, XGBoost can compute this exactly (*pred_contribs*), not as a slow approximation. For the worst band, those contributions are in decibels of predicted margin and add back up to the prediction. Green / positive means “this knob, at this setting, is adding headroom”. Red / negative means it is taking headroom away. The chart is sorted by absolute value so you can see the large movers first, regardless of sign.

The $\pm$ band on the score

Five extra margin models were trained on bootstrap resamples of the same simulated data. The $\pm$ you see is their disagreement, mapped onto the 0–100 score. It is a floor on uncertainty, not lab uncertainty. All five members went to the same school. They cannot tell you what the school got wrong about reality.

The path to real measurements

The file **calibrate.py** already exists. It is built for tens of chamber points, not thousands. It learns a small residual — measured margin minus simulated margin — as a function of the six design parameters, with the same monotone signs, and clamps it so a correction cannot invert physics. It has been tested on synthetic CSVs. It has **not** been run on chamber data. Until it has, every figure remains relative to the simulator, and the product says so on screen.

A few dozen real points would shift the absolute score toward lab reality, turn the $\pm$ into an empirical residual, and show which band the lumped cable model misplaces (likely 5–30 MHz). They would not replace the physics prior, and they would not turn this into a certification oracle.

PART V · THE MEETING

How to present this without getting hurt

Who is in the room

Person	What they care about	Lead with
Engineering manager	Schedule, lab cost, whether this reduces surprises	Pre-compliance gap. Demo ultra-rise versus a shorter cable. Do not open with XGBoost.
Drive / EMC engineer	Whether the physics is adult, whether you overclaim	Common-mode, CISPR RBW, limit-curve honesty, monotone constraints, design-only MAE.
Compliance / quality	Whether anyone will wave this as a certificate	The disclaimer, the PDF title, the missing signature block. Then the calibration hook.
Data / AI lead	Whether this is a demo wrapped around a black box	Synthetic labels, ablation versus full-model scores, SHAP identity, why spectral features are a cheat.
Executive	What we would buy, fund, or partner on	One-minute pitch. Then: chamber data is the unlock, not more UI.

A twelve-minute demo script

- **0:00 — Frame.** Say the one-minute pitch. Point at the disclaimer in the header. Do not skip this to “save time”.
- **1:30 — Pick Ultra-Rise.** “Eighty-five metres of motor cable in a tall shaft. This is the case that should look uncomfortable.”
- **2:30 — Parameters.** Flick dv/dt and shielding so they see they are ordinary design numbers. Mention the input-filter slider does not feed the EMC model.
- **4:00 — Results badge.** Read the framing sentence on the page out loud. Then the tier. Then the per-band margins. Never the badge alone.
- **6:00 — Why this margin.** Largest bar first. Cable will dominate. Shielding will likely be a large green bar. Say: “that is already working; it is why the recommendations may not tell you to add more shield.”
- **7:30 — What to change next.** Read the divergence sentence between the two panels. This is where trust is won or lost.
- **8:30 — Signals / THD.** Ten seconds on common-mode voltage versus motor PWM. “Different question, different timescale.”
- **9:30 — Compare.** Shorten the cable or drop dv/dt. Show the score move. Relative change is the product.
- **10:30 — Methodology.** Open it. Land on the limit-curve box and the design-only table. Close by asking for a chamber CSV, not for applause.

Questions you will get

Does this certify the drive?

No. Accredited measurement is the only certification path. This is a design-risk indicator against a synthetic envelope.

Can we skip the chamber if the badge is LOW?

No. A LOW result means the simulator is comfortable relative to our curve. It is a reason to walk into the lab calmer, not a reason to cancel the lab.

What is the accuracy?

Against our own physics, the design-only models are about 1.1–1.9 dB mean absolute error on margin, depending on the band. That is not accuracy against hardware, which is unmeasured. Do not quote 99% balanced accuracy from the full models. Those models can see the spectrum they are scored on.

Why machine learning if you already simulate?

Speed for sweeps, and a surface we can calibrate with sparse lab points without rewriting the simulator. The simulator remains the prior. We did not replace physics with a black box.

Why not a full finite-element / SPICE model?

Wrong job. Early design needs direction in seconds from a handful of drawing-office numbers. Lumped cable, first-order shield, and a trapezoidal inverter are enough for that, and they are honest about what they omit: fixture parasitics, connector details, cabinet layout, radiated paths.

Is the limit EN 12016?

No. EN 12016 is named because that is the lift-sector standard people will mention. The numerical table is not redistributable. Our curve is CISPR 11 Group 1 Class A shaped, with our own tapers, flagged everywhere as an assumption.

Your SHAP list and your recommendations disagree.

They should, sometimes. SHAP explains the current point, including factors already maxed out. Recommendations only suggest parameters with travel left. Shielding is the usual example.

Random PWM is supposed to be better.

Often, at high carriers where lines are resolved. In CISPR band B with a 3–16 kHz carrier, the 9 kHz receiver already smears the lines. Spreading does not reduce energy; max-hold then catches the extra peaks. The model reports about +1 dB, not a 5–10 dB win. Volunteer this; it is a sign you read the measurement, not a bug you hide.

What do you need from us?

A CSV of chamber results: the six design parameters, measured peak per band, and the limit you actually used. Tens of points, not thousands. That is enough to run the calibration path that is already written. Mentoring on whether the lumped cable story matches your fixtures is the second gift; it does not require us to pretend your data is in the training set.

Landmines

If you are tempted to say	Say this instead
"AI-powered certification."	"A physics-informed pre-compliance risk indicator."
"Trained on EMC data."	"Trained on 5 000 simulated designs. No chamber data yet."
"99% accurate."	"About 1–2 dB vs our simulator from design parameters alone. Lab error is unknown."
"The $\pm$ is the uncertainty."	"The $\pm$ is disagreement among models of the same simulator. It is a floor."
"EN 12016 compliant."	"Below our assumed CISPR-shaped envelope, which is not the standard table."
"It models the whole EMC problem."	"Conducted 150 kHz–30 MHz on the motor-cable path. Not radiated, not immunity."
"Input filter will fix EMC."	"Input filter moves 50 Hz THD. It is excluded from the EMC model on purpose."

What a serious next step looks like

- Replace the synthetic anchors with a limit line the company has the right to use, if they want named-standard scoring.
- Run `calibrate.py` on a few dozen chamber points from production-like cable lengths, including at least some 5–30 MHz fails.
- Keep monotone constraints. Do not fine-tune them away to chase a flattering residual.
- Report residual by band in public, including the ugly band.
- Leave certification where it belongs: in the lab report.

PART VI · REFERENCE

Numbers you can quote if asked

Item	Value
Training designs	5 000 simulated (3 500 / 750 / 750 split)
EMC sample rate	200 MS/s, 2 ¹⁴ samples, 6 dwell segments, 9 kHz RBW
PQ sample rate	200 kS/s, 2 ¹⁴ samples, 50 Hz THD
DC-link (model)	565 V
Cable capacitance	100 pF/m to earth
Shield	up to 26 dB insertion loss + HF roll-off
Model family	XGBoost classifier + regressor per band, monotone constraints
SHAP	Exact tree pred_contribs on design-only margin models
Ensemble	5 bootstrap members for the score ±
Score map	50 + 50 tanh(margin_dB / 8), then 0.5·mean + 0.5·worst
Tiers	LOW ≥ 70, MODERATE 40–69, HIGH < 40
Full-model margin MAE	0.13–0.55 dB vs simulator (sees the spectrum)
Design-only margin MAE	1.06 / 1.23 / 1.88 dB (A / B / C)
Seed-to-seed simulator scatter	~0.25 dB
Stack	Python FastAPI + React, reportlab PDFs

Glossary

CISPR / EN 55011. The family of emission measurement methods and industrial-equipment limits this curve is shaped like. CISPR 16-1-1 defines the 9 kHz receiver bandwidth used here.

Common mode (CM). The part of the voltage that is common to all motor phases versus earth. It drives conducted emissions on the cable screen / earth path.

Conducted emissions. Noise measured on the power leads, typically 150 kHz–30 MHz, using a LISN and an EMI receiver.

dBμV. Decibels relative to one microvolt. The unit on a conducted-emission plot. 0 dBμV = 1 μV across the LISN.

Design-only ablation. Retraining the models with the spectrum hidden, so they must infer EMC from the six knobs. The honest skill figure.

dv/dt. How fast the voltage changes at a switching edge, in volts per microsecond. Sets the high-frequency spectral corner.

EN 12016. Product-family EMC standard for lifts, escalators and moving walks (immunity side, with emission practice around it). Named often; its tables are not in this tool.

LISN. Line Impedance Stabilisation Network. Gives the measurement a defined 50 Ω view of the mains and dumps noise into the receiver.

Margin. Limit minus emission, in dB. Positive = simulated headroom. Negative = simulated exceedance of the assumed curve.

Monotone constraint. A hard rule on a boosted tree: this input may only push the prediction up, or only down. Here, the physics signs.

PWM. Pulse-width modulation. How the inverter fakes a smooth motor voltage with a stream of DC-link pulses.

RBW. Resolution bandwidth. How wide a frequency “bin” the receiver integrates. 9 kHz in this product.

SHAP. A method for splitting one prediction into per-input contributions that add up. Used here in dB of margin.

Surrogate model. A fast stand-in for a slower simulator, trained on the simulator’s own outputs.

THD. Total harmonic distortion. How much extra energy sits in harmonics of 50 Hz. A power-quality number, not an EMC-limit number.

VFD / VVVF. Variable-frequency (variable-voltage) drive. The inverter that runs the traction machine or escalator motor.

XGBoost. A library for gradient-boosted trees. The specific ML method in this product, chosen because constraints and exact SHAP are native.

What this document is not

It is not a substitute for EN 12016, CISPR 11, CISPR 16, or any accredited test report. It is not a characterisation of any manufacturer's shipped drive. Preset names are archetypes. Mentoring or industry review that informed the risk framing is not a product endorsement and is not validation against that organisation's hardware. The software remains a simulation.

This is a simulation-based pre-compliance risk assessment intended to support early-stage design decisions. It does not replace accredited EMC certification testing (for example per EN 12016), and it does not predict certification outcomes.

Regenerate this file from the repository with `python tools/company_briefing_pdf.py` inside the backend environment.